Some of the problems are in the textbook:
If you use the 6th edition:
Q1: Exercise 7.1 No. 57, Q2: Exercise 8.1 No.41, Q3: Exercise 8.2 No.16, Q4: Exercise 9.3 No.32, Q5a. Exercise 9.5 No.19, Q8: Exercise 17.4 No.9
If you use the 8th edition:
Q1: Exercise 7.1 No.62, Q2: Exercise 8.1 No.45, Q3: Exercise 8.2 No.18, Q4: Not in the 8th edition, Q5a. Exercise 9.5 No.19, Q8: Exercise 17.4 No.9
Note: << should've been <= (less or equal to)

IDE Introduction to Single-Variable Calculus II

Final Exam: 2024 Feb 13th 9:30-11:30

*Open text. Yet, solve them by yourself and do not discuss with others.

*Show explicitly intermediate steps to make it clear how you reach your solutions.

1. Use method of cylindrical shells to find the volume generated by the region bounded by the following curve rotating about the y-axis.

$$y = \cos\left(\frac{\pi x}{2}\right), \qquad y = 0, \qquad 0 \ll x \ll 1$$

2. Find the length of the curve: $y = \int_1^t \sqrt{x^3 - 1}dx\,, 1 \ll t \ll 4$

3. Find the area generated by curve, $y = \frac{1}{4}x^2 - \frac{1}{2}\ln(x)\,, 1 \ll x \ll 2$, which rotates around y-axis.

4. Find the orthogonal trajectories of the family of curves: $y = \frac{x}{1+kx}$

5. Solve the following two initial-value problems of 1st-order linear differential equation.

(1) $xy' = y + x^2 \sin(x)\,,\ y(\pi) = 0$

(2) $2xy' + y = 6x,\ x > 0,\ y(4) = 20$

6. For the second-order differential equation, $y'' + y = e^x + x^3$

(1) Find the general solution for its homogeneous equation

(2) Find the special solution for the inhomogeneous equation

(3) Solve this equation under the initial condition of $y(0) = 2, y'(0) = 0$

7. Prove the Euler's relation $e^{i\theta} = cos\theta + i\ sin\theta$ using Maclaurin series on $sin\theta$ and $cos\theta$

8 (BONUS PROBLEM). Using power series to solve the differential equatio

$y'' - xy' - y = 0, y(0) = 1, y'(0) = 0$